

```
top - 12:46:09 up 1:17, 0 user, load average: 0.01, 0.02, 0.00
Tasks: 35 total, 1 running, 34 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.5 us, 0.7 sy, 0.0 ni, 98.7 id, 0.0 wa, 0.0 hi, 0.2 si, 0.0 st
MiB Mem : 7947.4 total, 5446.4 free, 1347.5 used, 1394.2 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 6599.9 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+  COMMAND
  451 root        20   0 1793224  45880  32400 S   0.3   0.6   0:03.89 containerd
 1462 mmabith+   20   0  42.4g 686504  54736 S   0.3   8.4   0:18.11 node
    1 root        20   0   4324   3384   3100 S   0.0   0.0   0:00.06 bash
    9 syslog    20   0 222140   3764   2408 S   0.0   0.0   0:00.01 rsyslogd
   25 root        20   0  38768  28088  10984 S   0.0   0.3   0:00.97 python
   26 root        20   0   6196   2672   2432 S   0.0   0.0   0:00.00 logger
   67 root       10 -10  12024   4164   3052 S   0.0   0.1   0:00.01 sshd
  319 root        20   0 2022092  84396  61196 S   0.0   1.0   0:00.69 dockerd
  416 root        20   0 1234268  12364   8288 S   0.0   0.2   0:00.39 editor-proxy
  417 root        20   0  16328   6424   5500 S   0.0   0.1   0:00.01 sudo
  421 root        20   0 1226100  2676   1988 S   0.0   0.0   0:00.00 tmux-agent
  512 root       10 -10  13996   9768   8392 S   0.0   0.1   0:00.02 sshd
  588 root       10 -10  13996   9828   8456 S   0.0   0.1   0:00.03 sshd
  654 root        20   0   2696   1460   1364 S   0.0   0.0   0:00.00 sleep
  656 mmabith+   10 -10  14384   6548   4860 S   0.0   0.1   0:00.01 sshd
  658 mmabith+   10 -10  14256   6480   4824 S   0.0   0.1   0:00.02 sshd
  659 mmabith+   10 -10   7340   3820   3572 S   0.0   0.0   0:00.00 bash
  662 mmabith+   10 -10   7604   4564   3952 S   0.0   0.1   0:00.00 bash
  665 mmabith+   10 -10   7340   3796   3544 S   0.0   0.0   0:00.00 bash
  666 mmabith+   10 -10   7340   3692   3428 S   0.0   0.0   0:00.00 start-shell.sh
  667 mmabith+   10 -10   6880   3280   2980 S   0.0   0.0   0:00.00 tmux
  669 mmabith+   10 -10   7304   2776   2016 S   0.0   0.0   0:00.06 tmux
  670 mmabith+   10 -10   8796   5932   4060 S   0.0   0.1   0:00.08 bash
  716 root        20   0   9196   4004   3628 S   0.0   0.0   0:00.00 runuser
  717 mmabith+   20   0   4324   3336   3068 S   0.0   0.0   0:00.00 sh
```

```
mmabitha912@cloudshell:~$ uname
Linux
mmabitha912@cloudshell:~$ uptime
12:46:47 up 1:17, 0 user, load average: 0.00, 0.02, 0.00
mmabitha912@cloudshell:~$ hostname
cs-408522461355-default
mmabitha912@cloudshell:~$ free
              total        used        free      shared  buff/cache   available
Mem:           8138096       1379968       5576912          1148       1427704       6758128
Swap:              0              0              0

mmabitha912@cloudshell:~$ dmesg
dmesg: read kernel buffer failed: Operation not permitted
mmabitha912@cloudshell:~$
```

```
0.7% Tasks: 35, 65 thr, 0 kthr; 1 running
1.3% Load average: 0.00 0.01 0.00
Mem: 1.08G/7.76G Uptime: 01:18:31
Swap: 0K/0K

Main PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+  Command
    1 root        20   0  122.4g 686504  54736 S   0.3   8.4   0:18.11 /usr/bin/node /usr/bin/scripts/ctrl.sh sleep infinity
    9 syslog    20   0  216M  3764  2408 S   0.0   0.0   0:00.00 /usr/sbin/rsyslogd
   10 syslog    20   0  216M  3764  2408 S   0.0   0.0   0:00.00 /usr/sbin/rsyslogd
   11 syslog    20   0  216M  3764  2408 S   0.0   0.0   0:00.00 /usr/sbin/rsyslogd
   12 syslog    20   0  216M  3764  2408 S   0.0   0.0   0:00.00 /usr/sbin/rsyslogd
   25 root        20   0  38768  28088  10984 S   0.0   0.3   0:01.00 /usr/bin/python /usr/bin/supervisord -n -c /google/devshell/supervisord.conf --pidfile=/var/run/supervisor.pid
   26 root        20   0   6196   2672   2432 S   0.0   0.0   0:00.00 logger -t supervisord
   67 root       10 -10  12024   4164   3052 S   0.0   0.1   0:00.01 sshd: /usr/sbin/sshd -p 22 -o AuthorizedKeysFile=/etc/ssh/keys/authorized_keys [listener] 0 of 10-60 startups
  319 root        20   0 1974M  84396  61196 S   0.0   1.0   0:00.18 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
  390 root        20   0 1974M  84396  61196 S   0.0   1.0   0:00.10 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
  391 root        20   0 1974M  84396  61196 S   0.0   1.0   0:00.01 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
  392 root        20   0 1974M  84396  61196 S   0.0   1.0   0:00.09 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
  393 root        20   0 1974M  84396  61196 S   0.0   1.0   0:00.00 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
  414 root        20   0 1974M  84396  61196 S   0.0   1.0   0:00.01 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
  416 root        20   0 1205M  1364   8288 S   0.0   0.2   0:00.02 /google/devshell/editor/proxy/editor-proxy --static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell
  417 root        20   0 12328  4624  5500 S   0.0   0.1   0:00.01 sudo /google/devshell/tmux-agent
  418 root        20   0 1205M  1364   8288 S   0.0   0.2   0:00.06 /google/devshell/editor/proxy/editor-proxy --static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell
  419 root        20   0 1205M  1364   8288 S   0.0   0.2   0:00.00 /google/devshell/editor/proxy/editor-proxy --static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell
  420 root        20   0 1205M  1364   8288 S   0.0   0.2   0:00.04 /google/devshell/editor/proxy/editor-proxy --static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell
  421 root        20   0 1197M  2676   1988 S   0.0   0.0   0:00.00 /google/devshell/tmux-agent
  424 root        20   0 1197M  2676   1988 S   0.0   0.0   0:00.00 /google/devshell/tmux-agent
  425 root        20   0 1197M  2676   1988 S   0.0   0.0   0:00.00 /google/devshell/tmux-agent
  426 root        20   0 1197M  2676   1988 S   0.0   0.0   0:00.00 /google/devshell/tmux-agent
```

```
mmabitha912@cloudshell:~$ lsb_release
-bash: $'\247lsb_release': command not found
mmabitha912@cloudshell:~$ arch
x86_64
mmabitha912@cloudshell:~$
```

```

x86_64
mmabitha912@cloudshell:~$ pwd
/home/mmabitha912
mmabitha912@cloudshell:~$ ls
mlops README-cloudshell.txt
mmabitha912@cloudshell:~$ mkdir linuxpractice
mmabitha912@cloudshell:~$ ls
linuxpractice mlops README-cloudshell.txt
mmabitha912@cloudshell:~$ cd linuxpractice
mmabitha912@cloudshell:~/linuxpractice$ pwd
/home/mmabitha912/linuxpractice
mmabitha912@cloudshell:~/linuxpractice$ touch day2
mmabitha912@cloudshell:~/linuxpractice$ ls
day2
mmabitha912@cloudshell:~/linuxpractice$ touch day2session1 day2session2
mmabitha912@cloudshell:~/linuxpractice$ ls
day2 day2session1 day2session2
mmabitha912@cloudshell:~/linuxpractice$ cp day2session1 day2session2
mmabitha912@cloudshell:~/linuxpractice$ ls
day2 day2session1 day2session2
mmabitha912@cloudshell:~/linuxpractice$ mv day2session2 day2session1
mmabitha912@cloudshell:~/linuxpractice$ ls
day2 day2session1
mmabitha912@cloudshell:~/linuxpractice$ mkdir backup
mmabitha912@cloudshell:~/linuxpractice$ ls
backup day2 day2session1
mmabitha912@cloudshell:~/linuxpractice$ touch day2session12
mmabitha912@cloudshell:~/linuxpractice$ ls
backup day2 day2session1 day2session12
mmabitha912@cloudshell:~/linuxpractice$ rm day2session12
mmabitha912@cloudshell:~/linuxpractice$ ls
backup day2 day2session1

```

GNU nano 7.2

```

Linux is important
MLOps uses Linux
Cloud engineers use Linux
Docker works on Linux
Kubernetes uses Linux

```

```
mmabitha912@cloudshell:~/linuxpractice$ cat notes.txt
mlops :Linux is important
Linux is important
MLOps uses Linux
Cloud engineers use Linux
Docker works on Linux
mmabitha912@cloudshell:~/linuxpractice$ head notes.txt
Linux is important
MLOps uses Linux
Cloud engineers use Linux
Docker works on Linux
Kubernetes uses Linux
mmabitha912@cloudshell:~/linuxpractice$ tail notes.txt
Linux is important
MLOps uses Linux
Cloud engineers use Linux
Docker works on Linux
Kubernetes uses Linux
mmabitha912@cloudshell:~/linuxpractice$ head -2 notes.txt
Linux is important
MLOps uses Linux
mmabitha912@cloudshell:~/linuxpractice$ tail -2 notes.txt
Docker works on Linux
Kubernetes uses Linux
mmabitha912@cloudshell:~/linuxpractice$ vim notes.txt
mmabitha912@cloudshell:~/linuxpractice$ cat notes.txt
mlops :Linux is important
MLOps uses Linux
Cloud engineers use Linux
Docker works on Linux
Kubernetes uses Linux
mmabitha912@cloudshell:~/linuxpractice$
```



```

mmabitha912@cloudshell:~$ ^C
mmabitha912@cloudshell:~$ cd linuxpractice
mmabitha912@cloudshell:~/linuxpractice$ ls
backup day2 day2session1 notes.txt
mmabitha912@cloudshell:~/linuxpractice$ touch security.txt
mmabitha912@cloudshell:~/linuxpractice$ ls -l
total 12
drwxrwxr-x 2 mmabitha912 mmabitha912 4096 May 14 12:58 backup
-rw-rw-r-- 1 mmabitha912 mmabitha912 12 May 14 13:04 day2
-rw-rw-r-- 1 mmabitha912 mmabitha912 0 May 14 12:56 day2session1
-rw-rw-r-- 1 mmabitha912 mmabitha912 113 May 14 13:27 notes.txt
-rw-rw-r-- 1 mmabitha912 mmabitha912 0 May 15 01:07 security.txt
mmabitha912@cloudshell:~/linuxpractice$ chmod 777 security.txt
mmabitha912@cloudshell:~/linuxpractice$ ls
backup day2 day2session1 notes.txt security.txt
mmabitha912@cloudshell:~/linuxpractice$ ls -l
total 12
drwxrwxr-x 2 mmabitha912 mmabitha912 4096 May 14 12:58 backup
-rw-rw-r-- 1 mmabitha912 mmabitha912 12 May 14 13:04 day2
-rw-rw-r-- 1 mmabitha912 mmabitha912 0 May 14 12:56 day2session1
-rw-rw-r-- 1 mmabitha912 mmabitha912 113 May 14 13:27 notes.txt
-rwxrwxrwx 1 mmabitha912 mmabitha912 0 May 15 01:07 security.txt
mmabitha912@cloudshell:~/linuxpractice$ chmod 644 security.txt
mmabitha912@cloudshell:~/linuxpractice$ sudo chown root security.txt
mmabitha912@cloudshell:~/linuxpractice$ ls -l
total 12
drwxrwxr-x 2 mmabitha912 mmabitha912 4096 May 14 12:58 backup
-rw-rw-r-- 1 mmabitha912 mmabitha912 12 May 14 13:04 day2
-rw-rw-r-- 1 mmabitha912 mmabitha912 0 May 14 12:56 day2session1
-rw-rw-r-- 1 mmabitha912 mmabitha912 113 May 14 13:27 notes.txt
-rw-r--r-- 1 root mmabitha912 0 May 15 01:07 security.txt
mmabitha912@cloudshell:~/linuxpractice$ sudo chgrp root security.txt

```

```

mmabitha912@cloudshell:~/linuxpractice$ ls -l
total 12
drwxrwxr-x 2 mmabitha912 mmabitha912 4096 May 14 12:58 backup
-rw-rw-r-- 1 mmabitha912 mmabitha912 12 May 14 13:04 day2
-rw-rw-r-- 1 mmabitha912 mmabitha912 0 May 14 12:56 day2session1
-rw-rw-r-- 1 mmabitha912 mmabitha912 113 May 14 13:27 notes.txt
-rw-r--r-- 1 root mmabitha912 0 May 15 01:07 security.txt
mmabitha912@cloudshell:~/linuxpractice$ sudo chgrp root security.txt
mmabitha912@cloudshell:~/linuxpractice$ ls -l
total 12
drwxrwxr-x 2 mmabitha912 mmabitha912 4096 May 14 12:58 backup
-rw-rw-r-- 1 mmabitha912 mmabitha912 12 May 14 13:04 day2
-rw-rw-r-- 1 mmabitha912 mmabitha912 0 May 14 12:56 day2session1
-rw-rw-r-- 1 mmabitha912 mmabitha912 113 May 14 13:27 notes.txt
-rw-r--r-- 1 root root 0 May 15 01:07 security.txt
mmabitha912@cloudshell:~/linuxpractice$ umask
0002
mmabitha912@cloudshell:~/linuxpractice$ touch testmask.txt
mmabitha912@cloudshell:~/linuxpractice$ ls -l
total 12
drwxrwxr-x 2 mmabitha912 mmabitha912 4096 May 14 12:58 backup
-rw-rw-r-- 1 mmabitha912 mmabitha912 12 May 14 13:04 day2
-rw-rw-r-- 1 mmabitha912 mmabitha912 0 May 14 12:56 day2session1
-rw-rw-r-- 1 mmabitha912 mmabitha912 113 May 14 13:27 notes.txt
-rw-r--r-- 1 root root 0 May 15 01:07 security.txt
-rw-rw-r-- 1 mmabitha912 mmabitha912 0 May 15 01:10 testmask.txt
mmabitha912@cloudshell:~/linuxpractice$ stat security.txt
  File: security.txt
  Size: 0          Blocks: 0          IO Block: 4096   regular empty file
Device: 8,17      Inode: 131459        Links: 1
Access: (0644/-rw-r--r--)  Uid: (   0/   root)   Gid: (   0/   root)
Access: 2026-05-15 01:07:08.240339666 +0000

```

```

mmmabitha912@cloudshell:~/linuxpractice$ stat security.txt
  File: security.txt
  Size: 0                Blocks: 0          IO Block: 4096   regular empty file
Device: 8,17      Inode: 131459      Links: 1
Access: (0644/-rw-r--r--)  Uid: (    0/   root)   Gid: (    0/   root)
Access: 2026-05-15 01:07:08.240339666 +0000
Modify: 2026-05-15 01:07:08.240339666 +0000
Change: 2026-05-15 01:09:32.051877498 +0000
 Birth: 2026-05-15 01:07:08.240339666 +0000
mmmabitha912@cloudshell:~/linuxpractice$ setfacl -m u:mmabitha912:rwX security.txt
-bash: setfacl: command not found
mmmabitha912@cloudshell:~/linuxpractice$ lsattr security.txt
-----e----- security.txt
mmmabitha912@cloudshell:~/linuxpractice$ sudo chatter +i security.txt
mmmabitha912@cloudshell:~/linuxpractice$ sudo usermod -aG sudo username
usermod: user 'username' does not exist
mmmabitha912@cloudshell:~/linuxpractice$

```

```

    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
2: eth0@if7: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1460 qdisc noqueue state UP group default qlen 1000
    link/ether fe:98:47:b9:2b:ee brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 10.88.0.3/16 brd 10.88.255.255 scope global eth0
        valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1460 qdisc noqueue state DOWN group default
    link/ether 5e:7c:86:10:63:80 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
        valid_lft forever preferred_lft forever
mmmabitha912@cloudshell:~/linuxpractice$ ping google.com
PING google.com (142.251.12.139) 56(84) bytes of data:
64 bytes from se-in-f139.1e100.net (142.251.12.139): icmp_seq=1 ttl=114 time=0.866 ms
64 bytes from se-in-f139.1e100.net (142.251.12.139): icmp_seq=2 ttl=114 time=0.970 ms
64 bytes from se-in-f139.1e100.net (142.251.12.139): icmp_seq=3 ttl=114 time=0.602 ms
64 bytes from se-in-f139.1e100.net (142.251.12.139): icmp_seq=4 ttl=114 time=0.535 ms
64 bytes from se-in-f139.1e100.net (142.251.12.139): icmp_seq=5 ttl=114 time=0.715 ms
64 bytes from se-in-f139.1e100.net (142.251.12.139): icmp_seq=6 ttl=114 time=0.604 ms
64 bytes from se-in-f139.1e100.net (142.251.12.139): icmp_seq=7 ttl=114 time=0.564 ms
64 bytes from se-in-f139.1e100.net (142.251.12.139): icmp_seq=8 ttl=114 time=0.503 ms
64 bytes from se-in-f139.1e100.net (142.251.12.139): icmp_seq=9 ttl=114 time=0.573 ms
64 bytes from se-in-f139.1e100.net (142.251.12.139): icmp_seq=10 ttl=114 time=0.792 ms
64 bytes from se-in-f139.1e100.net (142.251.12.139): icmp_seq=11 ttl=114 time=0.613 ms
64 bytes from se-in-f139.1e100.net (142.251.12.139): icmp_seq=12 ttl=114 time=0.622 ms
64 bytes from se-in-f139.1e100.net (142.251.12.139): icmp_seq=13 ttl=114 time=0.586 ms
64 bytes from se-in-f139.1e100.net (142.251.12.139): icmp_seq=14 ttl=114 time=0.625 ms
64 bytes from se-in-f139.1e100.net (142.251.12.139): icmp_seq=15 ttl=114 time=0.616 ms
^C
--- google.com ping statistics ---
15 packets transmitted, 15 received, 0% packet loss, time 14298ms
rtt min/avg/max/mdev = 0.503/0.652/0.970/0.124 ms
mmmabitha912@cloudshell:~/linuxpractice$ netstat

```


Active Internet connections (w/o servers)						
Proto	Recv-Q	Send-Q	Local Address	Foreign Address	State	
tcp	0	0	cs-408522461355-d:59112	sd-in-f102.1e100.:https	ESTABLISHED	
tcp	0	0	localhost:51106	localhost:970	ESTABLISHED	
tcp	0	0	localhost:ssh	localhost:48870	ESTABLISHED	
tcp	0	0	cs-408522461355-d:44980	sj-in-f95.1e100.n:https	TIME_WAIT	
tcp	0	0	localhost:ssh	localhost:48860	ESTABLISHED	
tcp	0	0	localhost:48860	localhost:ssh	ESTABLISHED	
tcp	0	0	localhost:46672	localhost:ssh	TIME_WAIT	
tcp	0	0	localhost:971	localhost:41868	ESTABLISHED	
tcp	0	0	localhost:53428	localhost:971	ESTABLISHED	
tcp	0	0	localhost:41868	localhost:971	ESTABLISHED	
tcp	0	0	localhost:971	localhost:53428	ESTABLISHED	
tcp	0	0	cs-408522461355-d:50980	cdn-185-199-109-15:http	ESTABLISHED	
tcp	0	0	localhost:48870	localhost:ssh	ESTABLISHED	
tcp	0	0	localhost:50612	localhost:970	ESTABLISHED	
tcp6	0	0	cs-408522461355-def:980	10.95.145.12:34082	ESTABLISHED	
tcp6	0	0	localhost:970	localhost:50612	ESTABLISHED	
tcp6	0	0	cs-408522461355-def:980	10.95.145.12:34070	ESTABLISHED	
tcp6	0	0	localhost:970	localhost:51106	ESTABLISHED	
tcp6	0	0	cs-408522461355-def:980	10.95.145.12:52198	ESTABLISHED	
tcp6	0	334	cs-408522461355-def:980	10.95.147.12:43930	ESTABLISHED	
tcp6	0	0	cs-408522461355-def:980	10.95.147.12:49176	ESTABLISHED	
tcp6	0	0	cs-408522461355-def:980	10.95.147.12:34100	ESTABLISHED	
Active UNIX domain sockets (w/o servers)						
Proto	RefCnt	Flags	Type	State	I-Node	Path
unix	3	[]	STREAM	CONNECTED	54141	
unix	3	[]	STREAM	CONNECTED	56933	
unix	3	[]	STREAM	CONNECTED	57002	
unix	3	[]	STREAM	CONNECTED	55844	
unix	3	[]	STREAM	CONNECTED	55838	
unix	3	[]	STREAM	CONNECTED	29845	

unix	3	[]	STREAM	CONNECTED	28272	
unix	3	[]	STREAM	CONNECTED	28808	
unix	3	[]	STREAM	CONNECTED	55476	
unix	3	[]	STREAM	CONNECTED	55494	
unix	3	[]	STREAM	CONNECTED	55843	
unix	3	[]	STREAM	CONNECTED	29776	
unix	3	[]	STREAM	CONNECTED	28293	
unix	3	[]	STREAM	CONNECTED	28804	
unix	3	[]	STREAM	CONNECTED	55483	
unix	3	[]	STREAM	CONNECTED	56935	
unix	3	[]	STREAM	CONNECTED	28273	
unix	3	[]	STREAM	CONNECTED	55985	
unix	3	[]	STREAM	CONNECTED	55842	
unix	3	[]	STREAM	CONNECTED	29843	
unix	3	[]	STREAM	CONNECTED	26297	
unix	3	[]	STREAM	CONNECTED	56939	
unix	9	[]	DGRAM	CONNECTED	25294	/dev/log
unix	2	[]	DGRAM	CONNECTED	25787	
unix	3	[]	STREAM	CONNECTED	55477	
unix	3	[]	STREAM	CONNECTED	56937	
unix	3	[]	STREAM	CONNECTED	55983	
unix	3	[]	STREAM	CONNECTED	29844	
unix	3	[]	STREAM	CONNECTED	57000	
unix	3	[]	STREAM	CONNECTED	54165	
unix	3	[]	STREAM	CONNECTED	28276	
unix	3	[]	STREAM	CONNECTED	55988	
unix	3	[]	STREAM	CONNECTED	29774	
unix	3	[]	STREAM	CONNECTED	28746	
unix	3	[]	STREAM	CONNECTED	28810	
unix	3	[]	STREAM	CONNECTED	55482	
unix	3	[]	STREAM	CONNECTED	28744	
unix	3	[]	STREAM	CONNECTED	28278	

```

mmabitha912@cloudshell:~/linuxpractice$ ss -tln
Netid      State      Recv-Q     Send-Q      Local Address:Port      Peer Address:Port      Process
tcp        LISTEN     0           128         127.0.0.1:44993         0.0.0.0:*
tcp        LISTEN     0           511         0.0.0.0:971            0.0.0.0:*
tcp        LISTEN     0           128         0.0.0.0:22              0.0.0.0:*
tcp        LISTEN     0           511         127.0.0.1:43571         0.0.0.0:*
tcp        LISTEN     0           511         127.0.0.1:43883         0.0.0.0:*
tcp        LISTEN     0           511         127.0.0.1:46879         0.0.0.0:*
tcp        LISTEN     0           511         127.0.0.1:38407         0.0.0.0:*
tcp        LISTEN     0           5         127.0.0.1:8998          0.0.0.0:*
tcp        LISTEN     0           4096        *:900                    *:~*
tcp        LISTEN     0           4096        *:922                    *:~*
tcp        LISTEN     0           4096        *:970                    *:~*
tcp        LISTEN     0           4096        *:981                    *:~*
tcp        LISTEN     0           4096        *:980                    *:~*
tcp        LISTEN     0           128         [::]:22                  [::]:~*
mmabitha912@cloudshell:~/linuxpractice$ traceroute google.com
-bash: traceroute: command not found
mmabitha912@cloudshell:~/linuxpractice$ wget https://example.com
--2026-05-15 03:02:20-- https://example.com/
Resolving example.com (example.com)... 104.20.23.154, 172.66.147.243, 2606:4700:10:1ac42:93f3, ...
Connecting to example.com (example.com)[104.20.23.154]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: unspecified [text/html]
Saving to: 'index.html'

index.html          [ <> ] 528 --.-KB/s in 0s

```

```

mmabitha912@cloudshell:~/linuxpractice$ curl https://example.com
<doctype html><html lang="en"><head><title>Example Domain</title><meta name="viewport" content="width=device-width, initial-scale=1"><style>body{background:#eee;width:60vw;margin:15vh auto;font-family:system-ui,sans-serif}h1{font-size:1.5em}div{opacity:0.8}a:link,a:visited{color:#348}</style></head><body><div><h1>Example Domain</h1><p>This domain is for use in documentation examples without needing permission. Avoid use in operations.</p><p><a href="https://iana.org/domains/example">Learn more</a></p></div></body></html>
mmabitha912@cloudshell:~/linuxpractice$ nslookup google.com
Server:      169.254.169.254
Address:      169.254.169.254#53

Non-authoritative answer:
Name:   google.com
Address: 64.233.170.113
Name:   google.com
Address: 64.233.170.139
Name:   google.com
Address: 64.233.170.101
Name:   google.com
Address: 64.233.170.102
Name:   google.com
Address: 64.233.170.100
Name:   google.com
Address: 64.233.170.138
Name:   google.com
Address: 2404:6800:4003:c00::71
Name:   google.com
Address: 2404:6800:4003:c00::65
Name:   google.com
Address: 2404:6800:4003:c00::66
Name:   google.com
Address: 2404:6800:4003:c00::64
mmabitha912@cloudshell:~/linuxpractice$ dig google.com

```

```

Name:   google.com
Address: 2404:6800:4003:c00::66
Name:   google.com
Address: 2404:6800:4003:c00::64

mmabitha912@cloudshell:~/linuxpractice$ dig google.com

; <<>> DiG 9.18.39-0ubuntu0.24.04.3-Ubuntu <<>> google.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 60665
;; flags: qr rd ra; QUERY: 1, ANSWER: 6, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;google.com.                IN      A

;; ANSWER SECTION:
google.com.                288     IN      A      64.233.170.113
google.com.                288     IN      A      64.233.170.139
google.com.                288     IN      A      64.233.170.101
google.com.                288     IN      A      64.233.170.102
google.com.                288     IN      A      64.233.170.100
google.com.                288     IN      A      64.233.170.138

;; Query time: 2 msec
;; SERVER: 169.254.169.254#53 (169.254.169.254) (UDP)
;; WHEN: Fri May 15 03:02:52 UTC 2026
;; MSG SIZE rcvd: 135

mmabitha912@cloudshell:~/linuxpractice$

```

```
mmabitha912@cloudshell:~/linuxpractice$ sleep 300
^C
[1]- Stopped                  sleep 300
mmabitha912@cloudshell:~/linuxpractice$ ps
  PID TTY          TIME CMD
  1626 pts/3    00:00:00 bash
  2137 pts/3    00:00:00 sleep
  2142 pts/3    00:00:00 ps
mmabitha912@cloudshell:~/linuxpractice$ ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root           1        0  0 02:08 ?        00:00:00 /bin/bash /google/scripts/onrun.sh sleep infinity
syslog         9        1  0 02:08 ?        00:00:00 /usr/sbin/rsyslogd
root          25        1  0 02:08 ?        00:00:01 /usr/bin/python /usr/bin/supervisord -n -c /google/devshell/supervisord.conf --pidfile=/var/run/supervisor.pid --logfile=/var/
root          26        1  0 02:08 ?        00:00:00 logger -t supervisord
root           67        1  0 02:08 ?        00:00:00 sshd: /usr/sbin/sshd -p 22 -o AuthorizedKeysFile=/etc/ssh/keys/authorized_keys [listener] 0 of 10-60 startups
root          313        1  0 02:08 ?        00:00:00 /usr/bin/dockerd -p /var/run/docker.pid --atu=1460 --registry-mirror=https://asia-mirror.gcr.io
root          411       25  0 02:08 ?        00:00:00 /google/devshell/editor/proxy/editor-proxy --static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/static-content
root          412       25  0 02:08 ?        00:00:00 sudo /google/devshell/tmux-agent
root          417       412  0 02:08 ?        00:00:00 /google/devshell/tmux-agent
root          454       313  0 02:08 ?        00:00:05 containerd --config /var/run/docker/containerd/containerd.toml
root          649        1  0 02:08 ?        00:00:00 sleep infinity
root          653       411  0 02:08 ?        00:00:00 runuser -l mmabitha912 -c export GOPATH=/home/mmabitha912/gopath:/google/gopath HOME=/home/mmabitha912 CLOUD_SHELL=true EDITOR
mmabith+      654       653  0 02:08 ?        00:00:00 sh /bin/code-oss-cloudshell --host=0.0.0.0 --port=971
mmabith+      675       654  0 02:08 ?        00:00:09 /google/devshell/editor/code-oss-for-cloud-shell/node /google/devshell/editor/code-oss-for-cloud-shell/out/server-main.js --ho
mmabith+      694        1  0 02:08 ?        00:00:00 tmux new-session -A -D -n cloudshell -s 69407086
mmabith+      695       694  0 02:08 pts/2    00:00:00 -bash
root          1611       67  0 02:57 ?        00:00:00 sshd: mmabitha912 [priv]
root          1613       67  0 02:57 ?        00:00:00 sshd: mmabitha912 [priv]
mmabith+     1615      1611  0 02:57 ?        00:00:00 sshd: mmabitha912@pts/0
mmabith+     1616      1613  0 02:57 ?        00:00:00 sshd: mmabitha912@pts/1
mmabith+     1617      1615  0 02:57 pts/0    00:00:00 bash -c (sudo touch /var/run/google/devshell/44993 &)          trap "sudo rm -f /var/run/google/devshell/44993" EXIT HUP IN
```

```
mmabith+     1617      1615  0 02:57 pts/0    00:00:00 bash -c (sudo touch /var/run/google/devshell/44993 &)          trap "sudo rm -f /var/run/google/devshell/44993" EXIT HUP IN
mmabith+     1620      1617  0 02:57 pts/0    00:00:00 /bin/bash --nrc --noprofile
mmabith+     1623      1616  0 02:57 pts/1    00:00:00 bash -c if [ -f /google/devshell/start-shell.sh ]; then /google/devshell/start-shell.sh '' '' '' '697407086' false else
mmabith+     1624      1623  0 02:57 pts/1    00:00:00 /bin/bash /google/devshell/start-shell.sh 697407086 false
mmabith+     1625      1624  0 02:57 pts/1    00:00:00 tmux new-session -A -D -n cloudshell -s 697407086
mmabith+     1626      694  0 02:57 pts/3    00:00:00 -bash
mmabith+     1995       675  4 02:57 ?        00:00:36 /google/devshell/editor/code-oss-for-cloud-shell/node --dns-result-order=ipv4first /google/devshell/editor/code-oss-for-cloud-
mmabith+     2006       675  0 02:57 ?        00:00:00 /google/devshell/editor/code-oss-for-cloud-shell/node /google/devshell/editor/code-oss-for-cloud-shell/out/bootstrap-for-k --ty
mmabith+     2026      1995  0 02:59 ?        00:00:00 /google/devshell/editor/code-oss-for-cloud-shell/node /google/devshell/editor/code-oss-for-cloud-shell/extensions/json-langua
mmabith+     2034      1995  1 02:59 ?        00:00:09 /google/devshell/editor/code-oss-for-cloud-shell/node /google/devshell/editor/code-oss-for-cloud-shell/extensions/geminicod
mmabith+     2041      1995  1 02:59 ?        00:00:10 /google/devshell/editor/code-oss-for-cloud-shell/node /google/devshell/editor/code-oss-for-cloud-shell/extensions/cloudcode/di
mmabith+     2065      1995  0 02:59 ?        00:00:00 /home/mmabitha912/.cache/cloud-code/cloudcode_cil/cloudcode_cil/fba419f4/cloudcode_cil duet -trace -logtostderr
mmabith+     2137      1626  0 03:10 pts/3    00:00:00 sleep 300
mmabith+     2145      1626  0 03:11 pts/3    00:00:00 ps -ef
mmabitha912@cloudshell:~/linuxpractice$ pgrep sleep
649
2137
mmabitha912@cloudshell:~/linuxpractice$ ps
  PID TTY          TIME CMD
  1626 pts/3    00:00:00 bash
  2137 pts/3    00:00:00 sleep
  2167 pts/3    00:00:00 sleep
  2178 pts/3    00:00:00 ps
mmabitha912@cloudshell:~/linuxpractice$ ps
  PID TTY          TIME CMD
  1626 pts/3    00:00:00 bash
  2137 pts/3    00:00:00 sleep
  2167 pts/3    00:00:00 sleep
  2178 pts/3    00:00:00 ps
mmabitha912@cloudshell:~/linuxpractice$ pgrep sleep
649
2137
2167
mmabitha912@cloudshell:~/linuxpractice$ kill 12345
-bash: kill: (12345) - No such process
mmabitha912@cloudshell:~/linuxpractice$ ^C
mmabitha912@cloudshell:~/linuxpractice$ ^C
mmabitha912@cloudshell:~/linuxpractice$ kill 649
-bash: kill: (649) - Operation not permitted
mmabitha912@cloudshell:~/linuxpractice$ ^C
mmabitha912@cloudshell:~/linuxpractice$ sleep 500 &
[3] 2202
[2]  Done                  sleep 300
mmabitha912@cloudshell:~/linuxpractice$ pgrep sleep
649
2137
2202
```

```
mmabitha912@cloudshell:~/linuxpractice$ jobs
[1]-  Stopped                  sleep 300    (wd: ~/linuxpractice)
[2]+  Stopped                  sleep 300
mmabitha912@cloudshell:~/linuxpractice$ bg
[2]+  sleep 300 &
mmabitha912@cloudshell:~/linuxpractice$ jobs
[1]+  Stopped                  sleep 300    (wd: ~/linuxpractice)
[2]-  Running                  sleep 300 &
mmabitha912@cloudshell:~/linuxpractice$ ps
  PID TTY          TIME CMD
  1626 pts/3    00:00:00 bash
  2137 pts/3    00:00:00 sleep
  2167 pts/3    00:00:00 sleep
  2178 pts/3    00:00:00 ps
mmabitha912@cloudshell:~/linuxpractice$ pgrep sleep
649
2137
2167
mmabitha912@cloudshell:~/linuxpractice$ kill 12345
-bash: kill: (12345) - No such process
mmabitha912@cloudshell:~/linuxpractice$ ^C
mmabitha912@cloudshell:~/linuxpractice$ ^C
mmabitha912@cloudshell:~/linuxpractice$ kill 649
-bash: kill: (649) - Operation not permitted
mmabitha912@cloudshell:~/linuxpractice$ ^C
mmabitha912@cloudshell:~/linuxpractice$ sleep 500 &
[3] 2202
[2]  Done                  sleep 300
mmabitha912@cloudshell:~/linuxpractice$ pgrep sleep
649
2137
2202
```



```

mmabitha912@cloudshell:~$ pgrep sleep
649
2137
2202
mmabitha912@cloudshell:~$ kill 3456
-bash: kill: (3456) - No such process
mmabitha912@cloudshell:~$ ^C
mmabitha912@cloudshell:~$ kill 2202
[3]- Terminated sleep 500
mmabitha912@cloudshell:~$ pgrep sleep
649
2137
mmabitha912@cloudshell:~$ sleep 500 &
[2] 2219
mmabitha912@cloudshell:~$ kill 3000
-bash: kill: (3000) - No such process
mmabitha912@cloudshell:~$ █

```

```

mmabitha912@cloudshell:~$ sleep 500 &
[3] 2202
[2] Done sleep 300
mmabitha912@cloudshell:~$ pgrep sleep
649
2137
2202
mmabitha912@cloudshell:~$ kill 3456
-bash: kill: (3456) - No such process
mmabitha912@cloudshell:~$ ^C
mmabitha912@cloudshell:~$ kill 2202
[3]- Terminated sleep 500
mmabitha912@cloudshell:~$ pgrep sleep
649
2137
mmabitha912@cloudshell:~$ sleep 500 &
[2] 2219
mmabitha912@cloudshell:~$ kill 3000
-bash: kill: (3000) - No such process
mmabitha912@cloudshell:~$ killall sleep
sleep(649): Operation not permitted
[2]- Terminated sleep 500
mmabitha912@cloudshell:~$ sleep 300
^Z
[2]+ Stopped sleep 300
mmabitha912@cloudshell:~$ fg
sleep 300
^Z
[2]+ Stopped sleep 300
mmabitha912@cloudshell:~$ jobs
[1]- Stopped sleep 300 (wd: ~/linuxpractice)
[2]+ Stopped sleep 300

```

```

mmabitha912@cloudshell:~$ jobs
[1]-  Stopped                  sleep 300  (wd: ~/linuxpractice)
[2]+  Stopped                  sleep 300
mmabitha912@cloudshell:~$ nice -n 10 sleep 300
^Z
[3]+  Stopped                  nice -n 10 sleep 300
mmabitha912@cloudshell:~$ pgrep sleep
649
2137
2229
2238
mmabitha912@cloudshell:~$ renice 5 -p 12345
renice: failed to get priority for 12345 (process ID): No such process
mmabitha912@cloudshell:~$

```

```

mmabitha912@cloudshell:~$ df
Filesystem                1K-blocks    Used Available Use% Mounted on
overlay                   98831908 76306364  22509160   78% /
tmpfs                     65536      0      65536    0% /dev
/dev/sdal                 98831908 76306364  22509160   78% /root
/dev/disk/by-id/google-home-part1 5018272  154992  4584804    4% /home
/dev/root                 2019696 1288424   731272   64% /usr/lib/modules
shm                       65536      0      65536    0% /dev/shm
mmabitha912@cloudshell:~$ du -h
12K      ./docker
4.0K     ./config/htop
8.0K     ./config/powershell
4.0K     ./config/procps
20K      ./config
96K      ./npm/_logs
100K     ./npm
8.0K     ./gemini/history/mmabitha912
8.0K     ./gemini/history/code-oss-for-cloud-shell/.git/logs/refs/heads
12K      ./gemini/history/code-oss-for-cloud-shell/.git/logs/refs
20K      ./gemini/history/code-oss-for-cloud-shell/.git/logs
8.0K     ./gemini/history/code-oss-for-cloud-shell/.git/objects/f2
4.0K     ./gemini/history/code-oss-for-cloud-shell/.git/objects/pack
4.0K     ./gemini/history/code-oss-for-cloud-shell/.git/objects/info
8.0K     ./gemini/history/code-oss-for-cloud-shell/.git/objects/4b
28K      ./gemini/history/code-oss-for-cloud-shell/.git/objects
8.0K     ./gemini/history/code-oss-for-cloud-shell/.git/refs/heads
4.0K     ./gemini/history/code-oss-for-cloud-shell/.git/refs/tags
16K      ./gemini/history/code-oss-for-cloud-shell/.git/refs
4.0K     ./gemini/history/code-oss-for-cloud-shell/.git/branches
8.0K     ./gemini/history/code-oss-for-cloud-shell/.git/info
68K      ./gemini/history/code-oss-for-cloud-shell/.git/hooks
168K     ./gemini/history/code-oss-for-cloud-shell/.git

```

```

8.0K     ./codeoss/extensions
4.9M     ./codeoss
152M     .
mmabitha912@cloudshell:~$ sudo fdisk -l
sudo: fdisk: command not found
mmabitha912@cloudshell:~$ ^C
mmabitha912@cloudshell:~$ sudo apt update
Get:1 https://cli.github.com/packages stable InRelease [3,917 B]
Get:2 https://apt.postgresql.org/pub/repos/apt noble-pgdg InRelease [180 kB]
Get:3 https://apt.postgresql.org/pub/repos/apt noble-pgdg/main amd64 Packages [778 kB]
Get:4 https://download.docker.com/linux/ubuntu noble InRelease [48.5 kB]
Get:5 https://packages.cloud.google.com/apt gcsfuse-noble InRelease [1,227 B]
Get:6 https://download.docker.com/linux/ubuntu noble/stable amd64 Packages [64.9 kB]
Get:7 https://packages.cloud.google.com/apt cloud-sdk InRelease [1,620 B]
Get:8 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:9 http://archive.ubuntu.com/ubuntu noble InRelease
Get:10 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:11 https://packages.cloud.google.com/apt cloud-sdk/main all Packages [2,006 kB]
Get:12 https://repo.mongodb.org/apt/ubuntu noble/mongodb-org/8.0 InRelease [3,005 B]
Get:13 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1,504 kB]
Get:14 https://packages.cloud.google.com/apt cloud-sdk/main amd64 Packages [4,716 kB]
Get:15 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [2,038 kB]
Get:16 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:17 https://repo.mongodb.org/apt/ubuntu noble/mongodb-org/8.0/multiverse amd64 Packages [73.3 kB]
Hit:18 https://ppa.launchpadcontent.net/dotnet/backports/ubuntu noble InRelease
Get:19 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [2,445 kB]
Get:20 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [2,147 kB]
Get:21 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [3,889 kB]
Fetched 20.3 MB in 7s (2,770 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done

```